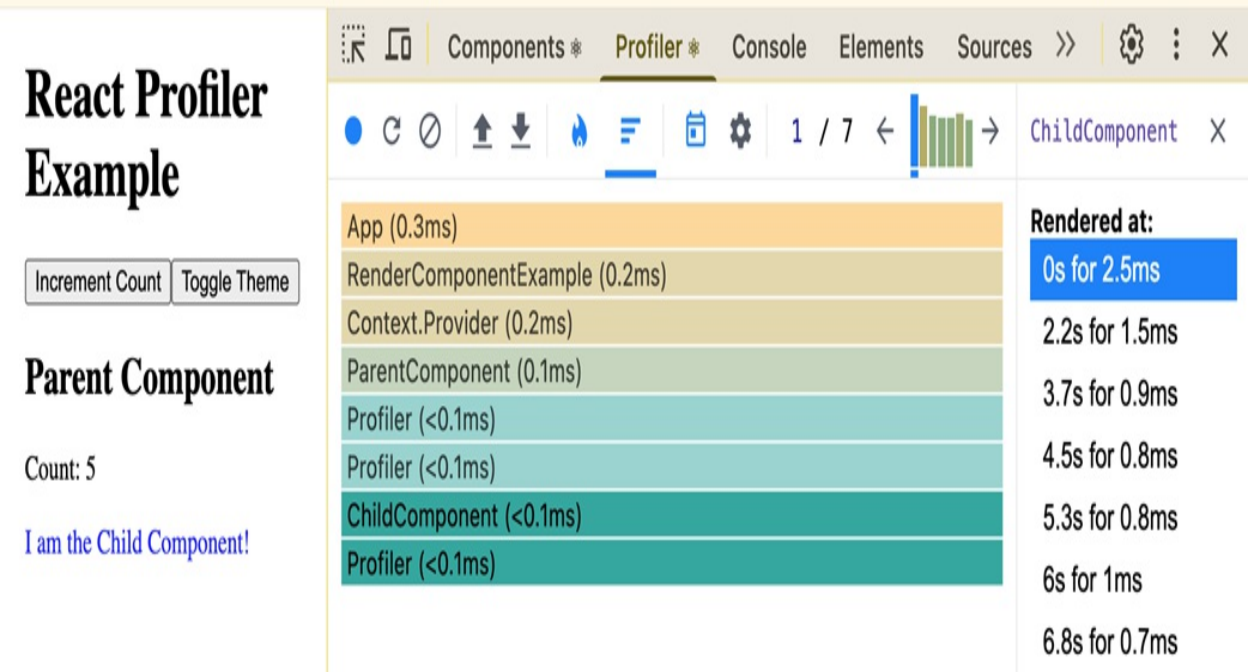
Figure 4.12 User interface and Profiler tab view in React Developer Tools.



The Profiler API is another valuable debugging tool. To demonstrate its functionality, the example in `RenderComponentExample.tsx` (listing 4.9) wraps relevant components within the `Profiler` component, enabling programmatic profiling of a specific React component subtree. The `Profiler` uses a callback function invoked on every render, providing detailed insights into performance metrics:

- `id`—The ID of the profiled component tree.
- `phase`—Indicates the rendering phase (e.g., "mount", "update", or "nested-update").
- `actualDuration`—The time spent rendering the component and its descendants during the current update.
- `baseDuration`—The estimated time to render the entire component tree without memoization.
- `startTime`—The timestamp when the render started.
- `commitTime`—The timestamp when the render was committed to the DOM.

Listing 4.10 shows the initial mount information about the `ChildComponent`.